

Name – Ganesh Malaji Ghule
Roll No - 37

Assignment No 8

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          Code 

Importing Necessary Libraries

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

Loading the dataset

```
In [2]: df=pd.read_csv('titanic.csv')

In [3]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column             Non-Null Count  Dtype
---  -
0   PassengerId         891 non-null    int64
1   Survived            891 non-null    int64
2   Pclass              891 non-null    int64
3   Name                891 non-null    object
4   Sex                 891 non-null    object
5   Age                 714 non-null    float64
6   SibSp               891 non-null    int64
7   Parch              891 non-null    int64
8   Ticket              891 non-null    object
```

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Code

```
5 Age 714 non-null float64
6 SibSp 891 non-null int64
7 Parch 891 non-null int64
8 Ticket 891 non-null object
9 Fare 891 non-null float64
10 Cabin 204 non-null object
11 Embarked 889 non-null object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

In [4]: df.describe()

Out[4]:
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [5]: df.shape
Out[5]: (891, 12)

Data Visualization

In [6]: sns.histplot(x='Fare',data=df)
```

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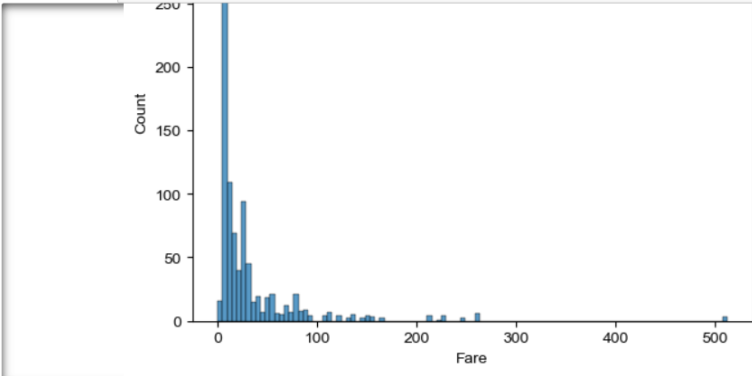
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Code

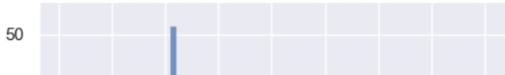
```
Data Visualization

In [6]: sns.histplot(x='Fare',data=df)
sns.set(rc={'figure.figsize':(3,3)})
```

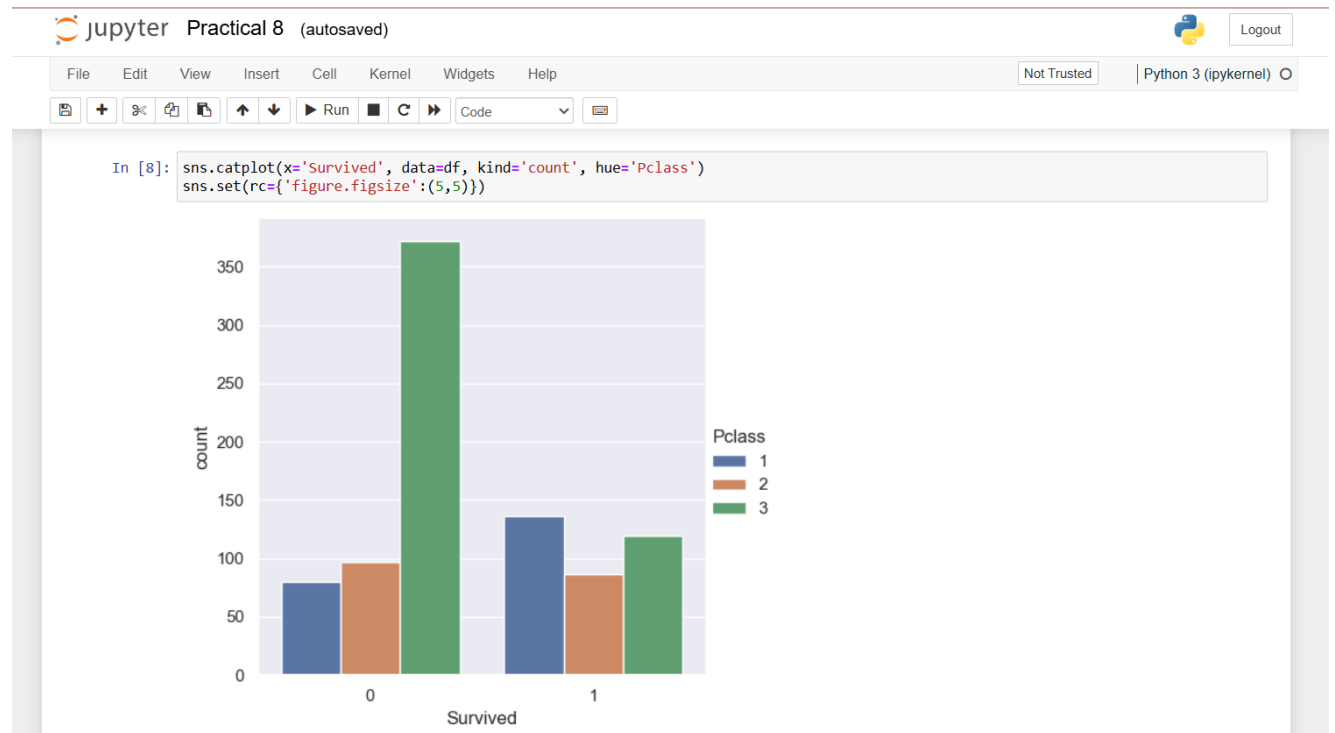


- This histogram shows how the price of the ticket for each passenger is distributed

```
In [7]: sns.displot(x='Age',data=df,bins=70)
sns.set(rc={'figure.figsize':(5,5)})
```



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